

Algebraic Fractions Homework

Name
Date
Course

Answers

$$\textcircled{1} \frac{2x}{5} + \frac{3x}{7}$$

$$\frac{29x}{35}$$

$$\textcircled{2} \left(-\frac{3e^5}{8}\right) \frac{4e^6}{5}$$

$$-\frac{3e^{11}}{10}$$

$$\textcircled{3} \frac{5z}{8} - \frac{3z}{4}$$

$$-\frac{z}{8}$$

$$\textcircled{4} \frac{5a^5}{7b^3} \left(-\frac{5a^6}{9b}\right)$$

$$-\frac{25a^{11}}{63b^4}$$

$$\textcircled{5} \frac{3x^2y^4}{7} \div \left(-\frac{y^2}{9x^4}\right)$$

$$-\frac{27x^6y^2}{7}$$

$$\textcircled{6} \frac{3c}{4} - \frac{c}{6}$$

$$\frac{7c}{12}$$

$$\textcircled{7} -\frac{2b^5}{5a} \left(\frac{4a^3}{b^2}\right)$$

$$-\frac{8ab^3}{5}$$

$$\textcircled{8} \left(-\frac{2r^2}{9s^2h}\right) \left(-\frac{5s^6}{12r^4h^3}\right)$$

$$\frac{5s^4}{54r^2h^4}$$

$$\textcircled{9} -\frac{7x}{8} + \frac{5x}{6}$$

$$-\frac{x}{24}$$

$$\textcircled{10} -\frac{5a^3b^9}{6c^4} \div \frac{3c^3b^4}{4a^6}$$

$$-\frac{10a^9b^5}{9c^7}$$

$$\textcircled{11} \frac{2y}{9} + \left(-\frac{5y}{6}\right)$$

$$-\frac{11y}{18}$$

$$\textcircled{12} -\frac{4z}{7} + \left(-\frac{z}{6}\right)$$

$$-\frac{31z}{42}$$

$$\textcircled{13} -\frac{3x}{8} - \frac{5x}{12}$$

$$-\frac{19x}{24}$$

$$\textcircled{14} -\frac{4a^3b^2}{7c^3} \left(-\frac{b^5}{c^4a^2}\right)$$

$$\frac{4ab^7}{7c^7}$$

$$\textcircled{15} -2z + \frac{2z}{3}$$

$$-\frac{4z}{3}$$

$$\textcircled{16} -\frac{2ed^3}{g^3} \div \left(-\frac{d^9}{8e^2g^6}\right)$$

$$\frac{16e^3g^3}{d^6}$$

$$\textcircled{17} \frac{5y}{8} - \left(-\frac{3y}{10}\right)$$

$$\frac{37y}{40}$$

$$\textcircled{18} -\frac{5k^9h^{12}}{9p^6} \div \left(-\frac{7k}{3hp^3}\right)$$

$$\frac{5h^{13}k^8}{21p^3}$$

$$\textcircled{19} -\frac{e}{2} (4e^3)$$

$$-2e^4$$

$$\textcircled{20} -6r^3g^5 \div \frac{3g^{10}}{4r^2}$$

$$-\frac{8r^5}{g^5}$$

$$\textcircled{21} -\frac{2z}{3} - \left(-\frac{5z}{6}\right)$$

$$\frac{z}{6}$$

$$\textcircled{22} (-3w^5) \frac{3}{2w^2}$$

$$-\frac{9w^3}{2}$$

$$\textcircled{23} \frac{y}{4} + \left(-\frac{3y}{8}\right) + \frac{7y}{16}$$

$$\frac{5y}{16}$$

$$(24) 2.7x + (-4.5x)$$

$$\boxed{-1.8x}$$

$$(25) \frac{2d^4}{7} \div (-5d^2)$$

$$\boxed{-\frac{2d^2}{35}}$$

$$(26) -\frac{5y}{6} + \frac{10y}{36} - (-\frac{11y}{12})$$

$$\boxed{\frac{13y}{36}}$$

$$(27) -\frac{2z}{15} - \frac{z}{10} - \frac{7z}{5}$$

$$\boxed{-\frac{49z}{30}}$$

$$(28) -1.9y + (-6.8y)$$

$$\boxed{-8.7y}$$

$$(29) 2a - \frac{4a}{5} + \frac{3a}{10}$$

$$\boxed{\frac{3a}{2}}$$

$$(30) -3z + 7.7z$$

$$\boxed{4.7z}$$

$$(31) \frac{2x}{3} + \frac{3x}{7}$$

$$\boxed{\frac{23x}{21}}$$

$$(32) (-\frac{3c^2}{7}) \frac{4c^8}{5}$$

$$\boxed{-\frac{12c^{10}}{35}}$$

$$(33) \frac{3y}{25} - \frac{2y}{5}$$

$$\boxed{-\frac{7y}{25}}$$

$$(34) \frac{2a^3}{3b^2} (-\frac{3a^4}{8b})$$

$$\boxed{-\frac{a^7}{4b^3}}$$

$$(35) -\frac{2c^2}{5d} (\frac{5d^3}{7c^4})$$

$$\boxed{-\frac{2d^2}{7c^2}}$$

$$(36) \frac{3e}{10} - \frac{7e}{15}$$

$$\boxed{-\frac{e}{6}}$$

$$(37) (-\frac{2x^2y^5}{7z^3}) (-\frac{5z^7y^6}{12x^8})$$

$$\boxed{\frac{5y^{11}z^4}{42x^6}}$$

$$(38) -\frac{h}{2} + \frac{5h}{8}$$

$$\boxed{\frac{h}{8}}$$

$$(39) \frac{c}{4} + (-\frac{c}{6})$$

$$\boxed{\frac{c}{12}}$$

$$(40) -\frac{2a^2bc^8}{7} (-\frac{b^3}{2c^5a^2})$$

$$\boxed{\frac{b^4c^3}{7}}$$

$$(41) 4e^5 (-\frac{2e^2}{3})$$

$$\boxed{-\frac{8e^7}{3}}$$

$$(42) -\frac{3x}{10} + (-\frac{5x}{6})$$

$$\boxed{-\frac{17x}{15}}$$

$$(43) -\frac{5y}{9} - \frac{5y}{8}$$

$$\boxed{-\frac{85y}{72}}$$

$$(44) \frac{2a^2b^5}{5} \div (-\frac{3b^3}{4a^5})$$

$$\boxed{-\frac{8a^7b^2}{15}}$$

$$(45) \frac{3}{5w^8} (-2w^5)$$

$$\boxed{-\frac{6}{5w^3}}$$

$$(46) \frac{5z}{6} - (-\frac{4z}{9})$$

$$\boxed{\frac{23z}{18}}$$

$$(47) \frac{3b}{4} - b$$

$$-\frac{b}{4}$$

$$(48) -\frac{3s^3r^5}{5h^3} \div \frac{8hr^2}{9s^6}$$

$$-\frac{27s^9r^3}{40h^4}$$

$$(49) -\frac{A}{6} - (-\frac{7A}{18})$$

$$\frac{2A}{9}$$

$$(50) \frac{5B}{4} + (-\frac{7B}{12}) + \frac{B}{6}$$

$$\frac{5B}{6}$$

$$(51) -\frac{3a^4b^3}{10c^5} \div (-\frac{2b^5c^3}{7a^2})$$

$$\frac{21a^6}{20b^2c^8}$$

$$(52) (-\frac{7x^5}{6yz}) \div (-\frac{9}{8xyz^5})$$

$$\frac{28x^6z^4}{27}$$

$$(53) -\frac{3c}{4} + \frac{c}{7} - (-\frac{9c}{28})$$

$$-\frac{2c}{7}$$

$$(54) -\frac{5a}{8} - \frac{a}{4} - \frac{3a}{20}$$

$$-\frac{41a}{40}$$

$$(55) -11.2z + (-0.4z)$$

$$-11.6z$$

$$(56) -4hg^8 \div \frac{2g^5}{3h^4}$$

$$-6h^5g^3$$

$$(57) \frac{2a}{3} - 3a + \frac{a}{2}$$

$$-\frac{11a}{6}$$

$$(58) 6y + (-9.5y)$$

$$-3.5y$$

$$(59) \frac{7d^3}{6} \div (-2d^8)$$

$$-\frac{7}{12d^5}$$

$$(60) -2.43x - 4.8x$$

$$-7.23x$$